NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-20/27E
12 August 1986
SUPERSEDING
MIL-C-20/27D
15 June 1984

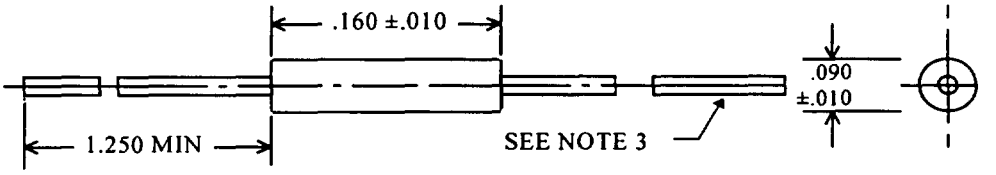
PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR75 AND CC75

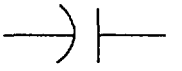
CC75 Inactive for design after 6 June 1984 use CCR75

This specification is approved for use by all Deparments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-PRF-20.



Inches	mm
.001	0.03
.005	0.13
.010	0.25
.016	.041
.090	2.29
.160	4.06
1.250	31.75



CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .016 (0.41 mm) +.005, -.001 (+0.13, -0.03 mm).

FIGURE 1. Styles CCR75 and CC75 capacitors.

Ⓔ denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: See table I.

Failure rate level (CCR75 only): M (1.0%), P (0.1%), R (.01%) or S (.001%).

Thermal shock and voltage conditioning (CCR75 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Ⓔ TABLE I. Capacitor characteristics.

Type designation 1/ 2/ 3/	Rated voltage volts, dc	Nom. cap. pF	Cap. tol.	Type designation 1/ 2/ 3/	Rated voltage volts, dc	Nom. cap. pF	Cap. tol.
CC-75CXR10--	200	0.10	B	CC-75CXR56--	200	0.56	B
CC-75CXR11--	"	0.11	"	CC-75CXR62--	"	0.62	"
CC-75CXR12--	"	0.12	"	CC-75CXR68--	"	0.68	"
CC-75CXR13--	"	0.13	"	CC-75CXR75--	"	0.75	"
CC-75CXR15--	"	0.15	"	CC-75CXR82--	"	0.82	"
CC-75CXR16--	"	0.16	"	CC-75CXR91--	"	0.91	"
CC-75CXR18--	"	0.18	"	CC-75CX1R0--	"	1.0	BC
CC-75CXR20--	"	0.20	"	CC-75CX1R1--	"	1.1	"
CC-75CXR22--	"	0.22	"	CC-75CX1R2--	"	1.2	"
CC-75CXR24--	"	0.24	"	CC-75CX1R3--	"	1.3	"
CC-75CXR27--	"	0.27	"	CC-75CX1R5--	"	1.5	"
CC-75CXR30--	"	0.30	"	CC-75CX1R6--	"	1.6	"
CC-75CXR33--	"	0.33	"	CC-75CX1R8--	"	1.8	"
CC-75CXR36--	"	0.36	"	CC-75CX2R0--	"	2.0	"
CC-75CXR39--	"	0.39	"	CC-75CK2R2--	"	2.2	"
CC-75CXR43--	"	0.43	"	CC-75CK2R4--	"	2.4	"
CC-75CXR47--	"	0.47	"	CC-75CK2R7--	"	2.7	BCD
CC-75CXR51--	"	0.51	"	CC-75CK3R0--	"	3.0	BCD

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

Type designation 1/ 2/	Rated voltage	Nom. cap.	Cap. tol.	Type designation 1/ 2/	Rated voltage	Nom. cap.	Cap. tol.
	volts, dc	pF			volts, dc	pF	
CC-75CK3R3--	200	3.3	BCD	CC-75CG510--	200	51	FGJ
CC-75CK3R6--	"	3.6	"	CC-75CG560--	"	56	"
CC-75CK3R9--	"	3.9	"	CC-75CG620--	"	62	"
CC-75CJ4R3--	"	4.3	"	CC-75CG680--	"	68	"
CC-75CJ4R7--	"	4.7	"	CC-75CG750--	"	75	"
CC-75CJ5R1--	"	5.1	"	CC-75CG820--	100	82	"
CC-75CJ5R6--	"	5.6	"	CC-75CG910--	"	91	"
CC-75CJ6R2--	"	6.2	"	CC-75CG101--	"	100	"
CC-75CJ6R8--	"	6.8	"	CC-75CG111--	"	110	"
CC-75CJ7R5--	"	7.5	"	CC-75CG121--	"	120	"
CC-75CH8R2--	"	8.2	"	CC-75CG131--	"	130	"
CC-75CH9R1--	"	9.1	"	CC-75CG151--	"	150	"
CC-75CH100-- 4/	"	10	FGJ	CC-75CG161--	"	160	"
CC-75CH110--	"	11	"	CC-75CG181--	"	180	"
CC-75CH120--	"	12	"	CC-75CG201--	"	200	"
CC-75CH130--	"	13	"	CC-75CG221--	"	220	"
CC-75CH150--	"	15	"	CC-75CG241--	"	240	"
CC-75CH160--	"	16	"	CC-75CG271--	50	270	"
CC-75CH180--	"	18	"	CC-75CG301--	"	300	"
CC-75CG200--	"	20	"	CC-75CG331--	"	330	"
CC-75CG220--	"	22	"	CC-75CG361--	"	360	"
CC-75CG240--	"	24	"	CC-75CG391--	"	390	"
CC-75CG270--	"	27	"	CC-75CG431--	"	430	"
CC-75CG300--	"	30	"	CC-75CG471--	"	470	"
CC-75CG330--	"	33	"	CC-75CG511--	"	510	"
CC-75CG360--	"	36	"	CC-75CG561--	"	560	"
CC-75CG390--	"	39	"	CC-75CG621--	"	620	"
CC-75CG430--	"	43	"	CC-75CG681--	"	680	"
CC-75CG470--	"	47	"				

1/ Complete type designation will include the following:

1st dash - Symbol "R" (for style CCR75) or dash will be deleted (for style CC75).

2nd dash - Applicable capacitance tolerance symbol.

3rd dash - Applicable failure rate level symbol (CCR75 only) or dash will be deleted (for style CC75).

(E) 2/ With respect to characteristics in previous issues of this specification sheet: for capacitance values of 2 pF or less, characteristic CX is interchangeable with characteristics CG and CH; for capacitance values of 2.2 pF through 3.9 pF, characteristic CK is interchangeable with characteristic CH; for capacitance values of 4.3 pF through 7.5 pF, characteristic CJ is interchangeable with characteristic CH; and for capacitance values of 10 pF through 18 pF, characteristic CH is interchangeable with characteristic CG. These parts shall be stocked under the same applicable NSN'S.

(E) 3/ Characteristic CX: Not practically measureable.

(E) 4/ Capacitance tolerance: ±1% (F) and ±2% (G).

Temperature coefficient and capacitance drift: In accordance with MIL-C-20. For capacitors with values of 18 pF or less, the temperature-coefficient tolerance shall be determined in accordance with table II.

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Ⓔ TABLE II. Temperature coefficient tolerances.

Temperature	Permissible capacitance change from capacitance at +25°C in ppm/°C				
	Characteristic				
	CX	CK	CJ	CH	CG
+125°C	1/	±250 ppm/°C	±120 ppm/°C	±60 ppm/°C	±30 ppm/°C
-55°C 2/	1/	+246.25 -326.25	+116.25 -166.25	+55.00 -91.25	+27.50 -53.75

1/ Not practically measurable.

2/ The ppm/°C values for -55°C were calculated by dividing ppm by negative 80°C.

Life: In accordance with MIL-C-20, operating condition 2.

Marking: Method I of MIL-STD-1285, as shown in following example:

JCK - "J" brand (ER only) (J); and characteristic (CK).

2R2 - Capacitance value.

CM6 - Cap. tolerance (C); FR level, CCR75 only (M); and year code (6 for 1986).

ATM - Lot code (A); and trademark (TM).

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - OS
Air Force - 85
DLA - ES

User activities:
Navy - AS, SH, MC

Preparing activity:
Air Force - 11

Agent:
DLA - ES

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